

Course Description | Whakamahuki

This course introduces “computational mechanical engineering”, including the computational modelling and analysis for applied mechanics and thermo-fluids, using Finite Element Analysis (FEA) and Finite Difference (FD) methods. These methods are based on computational approximations to the governing differential equations (ordinary and partial differential equations (PDEs)) that enable a rapid solution of models which have no analytical or straightforward solution.

This course translates what was learnt in applied mechanics (ENME202) and thermo-fluids (ENME215) into methods that can be used to solve more complex design and analysis problems. From this mathematical understanding of PDEs, students become familiar with the underlying principles of numerical solution techniques utilised in commonly-employed computational packages, e.g. COMSOL, used not in a “black box” manner, but with an appreciation of the underlying mathematics and numerical techniques embedded within them. This understanding of computational methods will be further augmented by the students’ own development and implementation of algorithms for numerical solutions of PDEs.

Learning Objectives | Hua Ako

- *Course topics:* with Learning Objectives (and Washington Accord (**WA**) and [UC Graduate Attributes](#)) identified;
- 1. *Bar (2DOF), Beam (4DOF), and Frame (6DOF) Elements: derivation, shape functions, assembly to models*
 - 1.1. Understand and apply the basic Finite Element Analysis (FEA) elements and formulations, including major method limitations (**WA1**)
- 2. *Extension to distributed loading and development of equivalent nodal loading*
 - 2.1. Recognise and apply different numerical solution terminology and techniques (**WA1**)
- 3. *Discussion on mathematical modelling and simulation*
 - 3.1. Appreciate properties and limitations of any numerical solution method (**WA1**)
 - 3.2. Understand computational strategies to maximize efficiency and minimize processing time. (**WA5, EIE3**)
- 4. *Introduction to Partial Differential Equations (PDEs), Separation of variables method and Fourier series*
 - 4.1. Recognise and classify the different types of PDEs (elliptic, parabolic and hyperbolic) (**WA1**)
 - 4.2. Use separation of variables solution method where applicable. (**WA1**)
- 5. *Generating a computational domain (CAD and mesh generation)*
 - 5.1. Able to code, MATLAB or similar, basic FEA assembly and analysis, and apply to solve problems (**WA5, EIE4**)
- 6. *Spatial discretisation: FD technique for elliptic problems, Time discretisation: application to parabolic PDEs*
 - 6.1. Understand the essential components of the PDE models for classical mechanical systems (**WA1**)
 - 6.2. Recognise & apply, Dirichlet and Neumann boundary conditions, for unsteady state and initial conditions (**WA1**)
- 7. *Important properties of numerical methods*
 - 7.1. Understand and apply d’Alembert solution and characteristics (**WA1**)
 - 7.2. Confidently apply standard analytic solution methods to classical PDEs used in mechanical analysis (**WA1**)
- 8. *Post-processing results / visualisation*
 - 8.1. Productively and confidently use generic computational packages (e.g. COMSOL) in the solution of “real world” problems in solid mechanics, fluid flow, and heat or mass transfer. (**WA5, EIE4**)
 - 8.2. Appreciate both the benefits and the limitations of such packages by comparison of numerical solutions with analytical solutions in situations where this is possible. (**WA5, EIE4**)

Course Structure

Lectures: 4 x 1 hr **lectures** weekly.

Computer Lab: 1 x 1.5 hr **computer laboratories** weekly.

Topics Covered in Weeks 1-4 with Professor Geoff Rodgers

Bar Elements (2DOF)

Derivation, Shape Functions, Assembly to Models, and examples.

Beam (4DOF) and Frame (6DOF) Elements

Derivation, Shape Functions, Assembly to Models, and examples. Extension to distributed loading and development of equivalent nodal loading.

Topics Covered in Weeks 5-12 with Dr James Hewett

Topics are for guidance only and may be subject to change.

- Discussion on mathematical modelling and simulation; Introduction to Partial Differential Equations (PDEs);
- The method of separation of variables and use of Fourier series to solve some PDEs;
- Generating a computational domain (CAD and mesh generation);
- Spatial discretisation, the Finite Difference technique for elliptic problems (applied to the steady heat equation);
- Time discretisation: application to parabolic PDEs (the transient heat diffusion equation);
- Hyperbolic PDEs (convection equation, wave equation);
- Important properties of numerical methods;
- Post-processing results / visualisation

Course Assessment | Aromatawai

More information is available from the [Course Information System \(CIS\) for ENME302-26S2](#).

Assessment	Date	% Final Grade
Quizzes	Weeks 6-10 (each Friday)	5%
Test 1	Week 6 – Tuesday 18 th August, 6:30 pm - 8:30 pm	25%
Assignment 1	Due in Week 7 – Monday 7 th September	10%
Assignment 2	Due in Week 12 – Tuesday 13 th October	10%
Final Examination	End-of-Year Examination Period	50%
Total Course Assessment		100%

A score of at least 40% in the Final Examination is required to pass this course, as well as achieving an overall score of at least 50% for the Total Course Assessment.

Texts and Readings | Tuhinga

Recommended reference texts:

James, Glyn, (2011) Advanced modern engineering mathematics. (4th ed.). Prentice Hall.

Patankar, Suhas V., (1980) Numerical heat transfer and fluid flow; Hemisphere Pub. Corp; McGraw-Hill.

Lecturer(s) and supporting staff details | Ingoa Kairuruku Akoranga

LECTURER:

GEOFF RODGERS (WEEKS 1-4)

ROOM: 5th Floor, Mech / Civil Building, Room E511
EMAIL: geoff.rodgers@canterbury.ac.nz
PHONE: +6433692270 – internal extension 92270

COURSE COORDINATOR & LECTURER:

JAMES HEWETT (WEEKS 5-12)

ROOM: 5th Floor, Mech / Civil Building, Room E547c
EMAIL: james.hewett@canterbury.ac.nz
PHONE: +6433690351 – internal extension 90351

Academic Policies (e.g. special consideration, dishonest practice) | Herenga Akoranga

The University of Canterbury and the [Department of Mechanical Engineering](#) has general policies that apply to all courses regarding such matters as dishonest practice, marks and grade boundaries, [late work](#), academic liaison, assistance for students with disabilities, reconsideration of grades, special considerations, and missing of tests etc.

Please consult the CIS ([Course Information System](#)) for more information. The Department reserves the right to refer dishonest practices to the University Proctor and where appropriate, to not mark the work. All information can be found on the [ENME-Undergraduate LEARN page](#).